

Симметризация Штейнера и изопериметрия

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Symmetrisation is a way of tidying a body up without changing its volume and without spoiling the quantity of interest. Together with Blaschke's selection theorem it yields complete proofs of the isodiametric and isoperimetric inequalities: first one shows that the ball is no worse, then one appeals to the existence of an extremal body.

Definition 1. For a convex body K and a unit vector u , the **Steiner symmetral** $S_u K$ is obtained by replacing every chord of K parallel to u by the segment of the same length centred on $u^\perp$.

Theorem 1 (Steiner 1838). $S_u K$ is convex, $|S_u K| = |K|$, and neither the diameter nor the surface area increases.

Theorem 2 (Bieberbach 1915). $|K| \leq \omega_n (\text{diam } K/2)^n$, with equality only for balls.

Theorem 3 (isoperimetric inequality in the plane). $L^2 \geq 4\pi A$ for a planar convex body of perimeter L and area A , with equality only for discs.

1. (Steiner 1838) Prove that $S_u K$ is convex and $|S_u K| = |K|$; prove that the diameter does not increase, and give an example showing that $S_u K \subseteq K$ is false in general.
2. (Bieberbach 1915) Deduce the isodiametric inequality. (*Hint: symmetrise in n orthogonal directions and note that for a centrally symmetric body $K \subseteq \frac{\text{diam } K}{2} B_2^n$.*)
3. Prove that Steiner symmetrization does not increase the perimeter of a planar convex body, and combine this with Blaschke's selection

theorem to prove Theorem 3. (*Hint: the extremal body coincides with all of its symmetrisations.*)

4. (classical) Among all quadrilaterals with given side lengths a, b, c, d (in this order), the cyclic one has the largest area; deduce Brahmagupta's formula $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ for it, where s is the semiperimeter.
5. (MexMAT, 2014) Let V be a convex polygon of area S and perimeter P . Prove that one can build a roof over V of volume at least $S^2/(2P)$, where a roof is the graph of a nonnegative 1-Lipschitz function vanishing on ∂V .
6. Prove that Steiner symmetrization does not increase the surface area of a convex body in $\mathbb{R}^n$. (*Hint: write the surface area as the integral of $\sqrt{1+|\nabla f|^2}$ over the upper and lower caps and apply the mean inequality.*)